

Ultrametric Root Counting Extended Abstract

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Let K be a complete field with respect to a non-archimedean discrete valuation v . We denote by $A = \{x \in K : v(x) \geq 0\}$ to the valuation ring of K , $\mathfrak{M} = \{x \in K : v(x) > 0\}$ to the maximal ideal of A , $\pi \in \mathfrak{M}$ to a generator of the ideal $\mathfrak{M}$ of A and $K_v = A/\mathfrak{M}$ to the residue field of K with respect to v . We assume that K_v is finite with q elements and characteristic p and that $v(\pi) = 1$. We also denote by v the unique extension of the valuation of K to its algebraic closure $\bar{K}$.

Let $f \in \mathbb{R}[X]$ be an univariate polynomial with real coefficients. Sturm's theorem allows us to determine the exact number of real roots of f in a given interval $[a, b]$. This is done by computing the difference between the number of sign changes of two sequences of real numbers called Sturm's sequences. We are interested in the analogue of Sturm's theorem over K , where K is a complete field with respect to non-archimedean discrete valuation. In this paper, we give an algorithmic method to compute the exact number of roots (total or with a given valuation) in K of a large class of polynomials in $K[X]$ called regular polynomials (see definition 1).

For any segment S of the Newton polygon of a regular polynomial $f \in K[X]$, the binomial containing the two terms of f corresponding to the vertices of S is called a lower binomial of f . We prove in theorem 2 that the number of roots of f in K^* coincides with the sum of the number of roots of all its lower binomials in K^* . Moreover, an explicit formula for the number of roots of each lower binomial is provided in theorem 3.

On the other hand, Descartes' rule of signs implies that any univariate polynomial $f \in \mathbb{R}[X]$ with exactly $t \geq 1$ monomial terms has at most $2t - 1$ real roots. Note that, Descartes' bound over the reals doesn't depend on the degree of the polynomial and it is linear in the number of monomial terms. In [5], H.W. Lenstra gave an analogue of Descartes' bound over the p -adic numbers. He showed that if $f \in K[X]$ is an univariate

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polynomial with coefficients in K , a finite extension of the p -adic rationals $\mathbb{Q}_p$, with at most $t \geq 1$ monomial terms, then the number of roots of f in K is $O(t^2(q-1)\log(t))$ where q is the cardinality of the residue field of K . As a consequence of our result of root counting, we can improve Lenstra's bound to $O(t(q-1))$ for regular polynomials.

Definition 1. A polynomial $f(X) = a_n X^n + a_{n-1} X^{n-1} + \dots + a_1 X + a_0 \in K[X]$ is regular if for any segment $S = (s, v(a_s)) \leftrightarrow (s', v(a_{s'}))$ of the Newton polygon (with $s > s'$), we have:

1. S does not contain any point from the set $\{(i, v(a_i)) : i = 1, \dots, n\}$ except the points $(s, v(a_s))$ and $(s', v(a_{s'}))$.
2. $p \nmid s - s'$ where $p = \text{char}(K_v)$.

The polynomial $a_s X^s + a_{s'} X^{s'}$ is called the lower binomial of f corresponding to the segment S .

Theorem 2. Let $f = a_n X^n + \dots + a_0 \in K[X]$ be a regular polynomial. Then the number of roots of f in K^* is equal to the sum of the number of roots of all its lower binomials in K^* .

Theorem 3. Let $f(X) = X^n + a_{n-1} X^{n-1} + \dots + a_1 X + a_0 \in A[X]$ with $p \nmid n$ and $a_0 \neq 0$. Write $l = v(a_0)$ and assume that $v(a_{n-i}) > il/n$ for all $i = 1, \dots, n-1$. Then the number R of roots of f in K^* is equal to the number of roots of the lower binomial $X^n + a_0$ in K^* . Moreover, if $n \nmid l$ we have $R = 0$, and if $n \mid l$ then

$$R = \begin{cases} \gcd(n, q-1) & \text{if } -\delta(a_0) \text{ is an } n\text{-th power in } K_v, \\ 0 & \text{otherwise.} \end{cases}$$

We also obtain in theorem 5 a reformulation of the classical Hensel's lifting. More precisely, we show a bijection between the set of roots of a monic polynomial (with coefficients in the valuation ring A of K) and some classes of roots of the reduced polynomial in $A/\mathfrak{M}^N$ where $\mathfrak{M}$ is the maximal ideal of A and N is a suitable positive integer. As a consequence, we prove that the number of roots in K of a polynomial in $K[X]$ with a given non-vanishing discriminant depends only on few "digits" of the coefficients (in corollary 6). It is easy to derive from the main result of this section an algorithm to compute the exact number of roots in K of a polynomial in $K[X]$.

Consider a monic polynomial $f(X) = X^n + a_{n-1} X^{n-1} + \dots + a_1 X + a_0 \in A[X]$. Assume that the discriminant $\Delta = \text{Res}_X(f, f')$ is non-zero and let $r = v(\Delta)$.

Definition 4. Let $Z_N \subseteq A/\pi^N A$ be the set of roots of f_N , where $f_N \in (A/\pi^N A)[X]$ denote the reduction of the polynomial f modulo π^N . Two roots $x, y \in Z_N$ are in the same cluster (denoted by $x \approx y$) if and only if either $x = y$ and $N \leq r$ or $x \equiv y \pmod{\pi^{r+1}}$ and $N > r$. The cluster containing a root $x \in Z_N$ is written $[x]$. The set of clusters is written $Z_N / \approx$.

Theorem 5. For any $N > 2r$, the number of roots of f in K is equal to $|Z_N / \approx|$. More precisely, the map $x \mapsto [x]$ is a bijection between the set of roots of f in A (or in K) and $Z_N / \approx$.

Corollary 6. Let $g = X^n + b_{n-1} X^{n-1} + \dots + b_0 \in A[X]$ be such that $v(a_i - b_i) > 2r$. Then f and g have the same number of roots in K .

References

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